

Karl-Gustav Peterson

Senior Full Stack & Mobile Engineer

SUMMARY

Senior Full-Stack Engineer with 10+ years of expertise in scalable microservices, composable architectures, and cross-platform mobile development across AWS, Azure, and GCP. Strong background in TypeScript, React, Next.js, Node.js, and React Native, delivering high-performance web and mobile applications with real-time capabilities and optimized Core Web Vitals. Experienced in event-driven systems (Kafka, RabbitMQ), advanced search architectures (Elasticsearch, Algolia), distributed caching (Redis), and Kubernetes-based deployments. Skilled in secure authentication, Stripe subscription systems, and AI integrations (OpenAI, Claude) for semantic enrichment and workflow automation. Proven track record of leading large-scale platform migrations and delivering production-grade systems in Agile environments.

PROFESSIONAL EXPERIENCE

Senior Full-Stack & Mobile Software Engineer

May 2022 – Jun 2026

Solvett | Warsaw, Poland

- Led migration of a 100K+ page medical platform to a composable architecture using **TypeScript**, **Next.js**, **React**, **Node.js**, **Sanity**, **Algolia**, **Elasticsearch**, **Redis**, **Docker**, **Kafka**, **RabbitMQ**, **Vercel**, and **AWS (S3, CloudFront, IAM, Route 53, EKS)**, integrating **Stripe** for subscription management.
- Architected scalable, multi-region content models in **Sanity Studio v3** (GROQ, custom schemas, RBAC, i18n, webhooks) enabling structured editorial workflows and automated publishing pipelines.
- Designed **REST** and **gRPC** backend services with **JWT** authentication and **RBAC**, leveraging **Kafka**, **RabbitMQ**, and **WebSocket** for event-driven processing and real-time communication, backed by optimized **PostgreSQL** schemas and indexing strategies.
- Implemented advanced search architecture with **Elasticsearch** and **Algolia**, building unified indexing pipelines, relevance tuning, and webhook-driven reindexing across 40K+ clinical articles and multimedia assets.
- Improved system performance through **Redis** caching, distributed session management, and optimized database queries under high traffic conditions.
- Containerized microservices using **Docker** and orchestrated with **Kubernetes (EKS)**, enabling horizontal scaling and zero-downtime deployments.
- Implemented hybrid **SSG**, **SSR**, and **ISR** using **Next.js App Router**, API Routes, and Edge Functions, deployed via **Vercel** and **AWS**, achieving green Core Web Vitals (LCP, INP, CLS).
- Optimized frontend performance using dynamic imports, **React.memo**, **Tailwind CSS**, **WebSocket**-driven live updates, and **CloudFront** edge caching.
- Engineered secure subscription flows with **Stripe API**, **Apple Pay**, and **Google Pay**, enforcing server-side validation, **AWS IAM** policies, and compliance-grade audit logging.
- Built cross-platform mobile applications with **React Native**, **Expo**, and **Redux Toolkit**, supporting CPD tracking, real-time updates via **WebSocket**, and offline-first synchronization.
- Integrated **GenAI** capabilities using **OpenAI GPT-4o** and **Anthropic Claude** APIs for clinical summarization, semantic enrichment, structured data extraction, and AI-assisted editorial workflows with structured output validation and guardrails.
- Implemented observability and monitoring with **Grafana**, centralized logging, and CI/CD automation via **Expo CLI**, **EAS Build**, and **GitHub Actions**, delivering 21 epics in a Scrum-based Agile environment.

Senior Mobile & Full-Stack Software Engineer

Aug 2020 – Apr 2022

Jelvix | Kharkiv, Ukraine

- Delivered a cross-platform mobile and web MVP using TypeScript, **React Native**, **Expo**, React.js, **Nest.js**, **Firebase**, **Sanity CMS**, and **Azure Cloud Services**.
- Built iOS and Android applications with structured state management and reusable UI components using React Native, Expo, **Redux Toolkit**, and **Styled Components**.
- Developed responsive frontend with React.js, TypeScript, **React Router**, and **Axios**, deployed via **Azure App Service** and **Azure Static Web Apps**.
- Engineered backend APIs and serverless business logic using Nest.js, **Firebase Cloud Functions**, and **Azure Functions** for scalable, modular, and maintainable microservices architecture.
- Implemented real-time data sync and secure storage using **Firebase Authentication**, **Firestore**, **Azure Blob Storage**, and security rules.
- Integrated dynamic content delivery with **Sanity CMS**, GROQ, **YouTube API**, optimized via **Azure CDN**.
- Built push notification workflows using **Expo Notifications**, **Firebase Cloud Messaging**, **Azure Notification Hubs**, and **Cloud Scheduler**.
- Implemented analytics via **Firebase Analytics**, **Azure Application Insights**, and custom event tracking.
- Automated lifecycle workflows using **Mailchimp API**, **HTTPS Functions**, and **Azure Logic Apps**.
- Configured CI/CD with **Expo EAS Build**, **GitHub Actions**, and **Azure DevOps Pipelines**.

CONTACTS

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EDUCATION

Bachelor's Degree in Computer Science

University of Tartu

Tartu, Estonia

Sep 2012 – Jun 2015

COMPETENCIES

- Full-Stack Development
- Cross-Platform Mobile Development
- Scalable Microservices & Distributed Systems
- Cloud-Native Architecture
- API Design & Development & Integration
- Event-Driven Architecture
- Search & Indexing
- Secure Authentication & Security
- Performance Optimization & Caching
- AI Integration & Workflow Automation
- Agile / Scrum Delivery

SKILLS

Backend & APIs

- Node.js, Nest.js
- REST APIs, GraphQL (Apollo Client/Server), gRPC
- Microservices architecture
- Event-driven systems (Kafka, RabbitMQ)
- Relational Databases: PostgreSQL, MySQL, Oracle (schema design, indexing)
- NoSQL Databases: AWS DynamoDB, MongoDB
- Serverless Backend services
- Redis (caching, distributed sessions)
- Automated testing
- Webhooks and event-driven pipelines

Mobile Software Engineer

Opus Online | Tartu, Estonia

Nov 2017 – Jul 2020

- Architected scalable cross-platform mobile applications using **React Native**, **Expo**, and **TypeScript**, implementing modular architecture with **React Navigation** and **Context API** for maintainability and rapid iteration.
- Engineered secure authentication flows including passwordless Magic Link, **JWT**, and **OAuth**, with secure storage via **Expo SecureStore** and role-based access control across **REST**, **gRPC**, and **GraphQL (Apollo Client)** services.
- Developed QR-based voucher scanning with **Expo Camera**, deep linking, background tasks, and real-time validation, ensuring low-latency and secure transaction processing.
- Optimized performance through efficient data-fetching strategies, intelligent **Apollo** caching, optimistic UI updates, memoization, lazy loading, and bundle optimization.
- Built reusable component libraries in **Storybook** following atomic design from **Figma**, improving UI consistency and development velocity.
- Established CI/CD pipelines with **GitHub Actions** and **Expo EAS Build**; managed releases via **App Store Connect** and **Google Play Console**, enforcing code quality with **ESLint**, **Prettier**, **Husky**, **Jest**, and **React Native Testing Library**.

Full Stack Software Engineer

Brocoders | Tallinn, Estonia

Jul 2015 – Oct 2017

- Designed and deployed a microservice-based onboarding application on **Google Cloud Platform (GCP)** using **Cloud Run** and **Cloud Build**, enabling scalable, containerized CI/CD-driven releases.
- Built the frontend with **Next.js**, **TypeScript**, **Bootstrap**, and **Storybook**, developing reusable component libraries and enforcing modular, maintainable architecture.
- Implemented secure API integrations without modifying the existing Amplience API, ensuring scalable and fault-tolerant cloud communication.
- Led structured **code reviews**, enforcing clean architecture, SOLID principles, and TypeScript best practices across the JavaScript codebase.
- Established automated testing pipelines with **Jest**, **Mocha**, and **React Testing Library**, integrating coverage checks, linting (**ESLint**, **Prettier**, **Husky**), and CI quality gates.
- Delivered iterative releases within Scrum using GCP-backed automated build, test, and deployment workflows.

PROJECTS

Healthcare & Medical Education Platform

<https://GNotebook.com>

Modernized and scaled GNotebook, a 100K+ page clinical reference platform, by migrating from a legacy CMS to a composable architecture built with React Native, Next.js, React, TypeScript, Node.js, Sanity, Algolia, Stripe, and Vercel, enabling high-performance delivery, multi-region content management, CPD compliance tracking, subscription monetization, and unified web and mobile experiences for medical professionals across the UK, US, Australia, and Ireland.

InsurTech Full MVP Platform

<https://veygo.com>

Led development of a cross-platform mobile and web MVP using React Native, Expo, React.js, TypeScript, Node.js, Firebase, and Sanity CMS to validate a new InsurTech learning product. The platform enabled progress tracking, personalized content delivery, push notifications, analytics-driven user behavior insights, and secure customer data handling to support rapid idea validation and risk mitigation.

Front-End Engineering

- TypeScript, JavaScript (ES6+)
- React.js, Next.js (SSR, SSG, ISR, App Router)
- React Native, Expo
- React Navigation, Context API, Redux Toolkit
- Tailwind CSS, Styled Components
- Storybook, Atomic Design
- WebSockets, Real-time UI updates
- Performance optimization (memoization, lazy loading, code splitting)

Mobile Development

- Cross-platform iOS & Android development
- App Store Connect & Google Play Console release management
- Push notifications (Expo, Firebase)
- Offline-first architecture

Authentication & Security

- JWT, OAuth 2.0
- Passwordless Magic Link
- RBAC (Role-Based Access Control)
- Secure storage (Expo SecureStore)
- API security & server-side validation
- Stripe subscription integration

Cloud & DevOps

- AWS (S3, CloudFront, IAM, Route 53, EKS)
- Azure (App Service, Static Web Apps, Functions, Blob Storage)
- Google Cloud Platform (GCP) (Cloud Run, Cloud Build)
- Docker, Kubernetes
- CI/CD (GitHub Actions, Expo EAS, Azure DevOps, GCP Cloud Build)
- Observability (Grafana, Application Insights)

CMS & Search Platforms

- Sanity CMS (v3) – GROQ, custom schemas, webhooks, i18n
- Elasticsearch
- Algolia
- Content modeling & indexing pipelines

Testing & Code Quality

- Jest, Mocha
- React Testing Library
- ESLint, Prettier, Husky
- Unit, integration, and API testing
- Code review standards & pull request governance

AI & Data Integration

- OpenAI (GPT-4o) integration
- Anthropic Claude integration
- Structured output validation
- Semantic enrichment & automation workflows